

LEAKLISTEN · DESIGN DECISIONS REGISTER

LeakListen design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/leaklisten		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions, items to confirm, value engineering and decisions made to date

LeakListen design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Design for construction as a whole: the neck bar hanger, flanged end plugs, top plug fittings, internal chassis and puck changes made under Amish's 2026-09-30 instruction	Accept, or ask for changes	Accept	Every component; nothing is bought or made before this is accepted	LKL-DDR-003, P1 to P9
2	R12 mass limit: the set is 1.35 kg with the neck bar against 1.0 kg	(a) restate R12 as 1.0 kg for the logger, cable and sensor (0.96 kg) and count the neck bar as site hardware; (b) relax R12 to 1.5 kg in all; (c) look for a lighter hanger (1 mm wall tube saves about 55 g; still over)	(a)	The first check that weighs the set	LKL-DDR-003, A1
3	Hanger for necks outside about 570 to 710 mm, or with poor walls	(a) neck bar as modelled, with a longer inner tube for wider necks; (b) a bracket on two concrete screws in the neck wall, with the utility's consent to drill	(a) for the prototype; confirm the pilot district's neck sizes before any deployment	Neck bar tubes and feet	LKL-DDR-003, A2
4	First partner utility and region for co-design and a pilot district	Any utility willing to co-design	None yet	Radio band of the antenna and board; the neck sizes of item 3	LKL-DDR-002, O1
5	Whether a vibration calibration check belongs in LeakListen or in CalRig	LeakListen; CalRig	None yet	Not part of the TRL 3 build; needed for the sensor checks at TRL 4	LKL-DDR-002, O2
6	Appearance model and product renders: status light pipe on the top plug, board orientation, cable route and render context (raised 2026-09-26)	Keep or drop each, as listed in the review note	Keep the light pipe as a brief blink at power-up only; accept the other three as appearance only	Top plug (one more hole if the light pipe is kept); the renders are redone on Amish's Mac	<code>docs/REVIEW.md</code> , 2026-09-26

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The pot magnet's stud is M6 and can be cut to 6 mm	The puck base is tapped M6 for 6 mm of thread	LKL-DDR-003, P7
2	The flat antenna's centre stud size, and that its lead ends in an SMA plug	Sets the bracket's stud hole and the top plug's connector	LKL-DDR-003, P2 and P4
3	The SMA bulkhead's panel hole and the M12 socket's thread (M16 x 1.5 assumed)	Sets the holes in the two end plugs	LKL-DDR-003, P4 and P6
4	The square tube inserts suit 1.5 mm wall tube and take an M10 levelling foot	The feet wedge the bar	LKL-DDR-003, P1
5	The main board fits 38 x 110 mm with its components within 8 mm	The chassis and the tube bore are sized to it	LKL-DDR-003, P5
6	The capacitor and cell sizes (16 x 30 mm, C size 26.2 x 50 mm)	The chassis clips are printed to them; the cell clears the bore by 0.8 mm	LKL-DDR-003, P5

■ Value engineering

Value-engineering target: USD 130 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 141 (USD 11 over the target). Main cost drivers and savings worth trying:

- The largest lines are the main board (USD 30, a prototype price; its design is TRL 4 work), the neck bar hanger (USD 18), the logger housing with its turned plugs (USD 16), the sensor cable (USD 12) and the primary cell (USD 12).
- Making the design constructable added USD 28 to the concept's USD 113: the neck bar (USD 12 more than the strap), the turned plugs, O-rings and screws (USD 4), the internal chassis (USD 4), the eye bolt and SMA bulkhead (USD 7) and the thicker puck (USD 1).
- Savings worth trying: a fixed-length bar cut to each district's neck width, with one foot and no inner tube or pin (about USD 4 and 80 g); plugs turned in batches or moulded later; a main board price that should fall once its design is done; a cable bought by the reel with a field-fit M12 plug.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	D1 to D6: nightly noise-level screening; flat antenna under the cover; piezo disc sensor with the hydrophone kept for plastic networks; reuse of the FieldNode STM32WL core; C-size cell; raw samples never leave the device	Amish: "i accept all your recommendations, go with them across all repos."	LKL-DDR-001, LKL-DDR-002
2026-09-25	D7: measure cover loss first, then a gateway within about 0.5 km of each district with iron covers	Amish, same instruction	LKL-DDR-002
2026-09-25	D8: the contact sensor's scope is metallic mains; the hydrophone variant answers plastic mains	Amish, same instruction	LKL-DDR-002
2026-09-25	D9: 42 mm pot magnet with a keeper plate	Amish, same instruction	LKL-DDR-002
2026-10-01	Budget treated as a value-engineering target, reported as over or under rather than met or not met	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	This register; LKL-CAL-001 v0.3